

Predictive Model Plan

1. Model Logic:

Predictive Model: Logistic Regression for Delinquency Prediction

The objective of the model is to predict whether a customer is likely to become delinquent based on their financial and credit-related information.

Key Input Features:

Payment History

Credit Utilization Ratio

Income Level

Debt-to-Income Ratio

Account Age

Model Process

Collect customer financial and credit data.

Prepare and organize the data for analysis.

Select the most relevant features influencing delinquency.

Train a Logistic Regression model using historical customer records.

Generate a probability score for each customer.

Classify customers into Low, Medium, or High-Risk categories.

Use the results to support lending and risk-management decisions.

Expected Output:

The model produces a delinquency risk score indicating the likelihood that a customer will miss future payments or become delinquent.

2. Justification for Model Choice:

I selected Logistic Regression because it is widely used in financial risk prediction and provides clear, explainable results. The model offers a good balance between accuracy and transparency, allowing analysts to understand how different factors influence delinquency risk. It is easy to implement, computationally efficient, and suitable for large customer datasets. Since financial institutions must often explain lending decisions to regulators and customers, Logistic Regression is an appropriate choice for Geldium's business needs. Although more complex models may achieve slightly higher accuracy, this model provides better interpretability and easier deployment.

3. Evaluation Strategy:

The model's performance will be evaluated using Accuracy, Precision, Recall, F1 Score, and AUC-ROC.

Accuracy: Measures the overall percentage of correct predictions.

Precision: Measures how many customers predicted as delinquent are actually delinquent.

Recall: Measures how many actual delinquent customers are correctly identified.

F1 Score: Provides a balanced measure of Precision and Recall.

AUC-ROC: Evaluates the model's ability to distinguish between delinquent and non-delinquent customers.

Bias Detection and Fairness:

Compare model performance across different customer groups.

Monitor false positive and false negative rates.

Regularly review feature importance to identify potential bias.

Ethical Considerations:

Customer data should be handled securely and responsibly. Predictions should support decision-making rather than automatically determining customer outcomes. The model should remain transparent, fair, and compliant with financial regulations.